MILITARY SPECIFICATION

COUPLING ASSEMBLIES, QUICK-DISCONNECT, SEXLESS TYPE

This specification is approved for use within the USA Belvoir Research, Development, and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for procurement of quick-disconnect coupling assemblies sexless type, used to provide a positive seal disconnect between components of an aircraft fuel servicing system.

1.2 Classification. Straight configuration or fitting configuration coupling assemblies shall be as specified (see 6.2).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications, and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATONS

FEDERAL

QQ-A-225/8	- Aluminum and Aluminum Alloy Bar, Rod, Wire or Special Shapes, Rolled, Drawn or Cold Finished.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4730

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:12Z
Check the source to verify that this is the current version before use.

QQ-S-763 - Steel Bars, Wire, Shapes-Forgings, Corrosion Resistant.
TT-S-735 - Standard Test Fluids, Hydrocarbon.
VV-F-800 - Fuel Diesel Oil.
PPP-B-601 - Boxes, Wood, Cleated Plywood.
PPP-B-636 - Box, Shipping, Fiberboard.
PPP-T-60 - Tape, Packaging, Waterproof.

MILITARY

MIL-P-116 - Preservation, Methods of.
MIL-B-121 - Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-P-5315 - Packing, Preformed, Hydrocarbon Fuel Resistant.
MIL-C-5541 - Chemical Conversion Coatings on Aluminum Alloys.
MIL-G-5572 - Gasoline Aviation, Grades 80/87, 100/130, 115/145.
MIL-T-5624 - Turbine Fuel, Aviation, Grades JP-4 and JP-5.
MIL-C-7024 - Calibrating Fluid, Aircraft Fuel System Components.
MIL-A-8625 - Anodic Coatings for Aluminum and Aluminum Alloys.
MIL-C-11866 - Castings, Precision, Nonferrous.
MIL-A-12545 - Aluminum Alloy Impacts.
MIL-R-25988 - Rubber, Fluorosilicone Elastomer, Oil-and-Fuel-Resistant, Sheets, Strips, Molded Parts, and Extruded Shapes.
MIL-C-27487 - Coupling Halves, Quick-Disconnect, Cam-Locking Type.
MIL-F-46162 - Fuel, Diesel, Referee Grade.
MIL-C-81975 - Coupling, Regulated, Aerial Pressure Refueling Type MA-3.
MIL-T-83133 - Turbine Fuel, Aviation, Kerosene Type, Grade JP-8.
MIL-W-83420 - Wire Rope, Flexible, for Aircraft Control.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-130 - Identification Marking of U.S. Military Property.
MIL-STD-810 - Enviornmental Test Methods and Engineering Guidelines.
MIL-STD-889 - Dissimilar Metals.
MIL-STD-1472 - Human Engineering Design Criteria for Military Systems, Equipment and Facilities.

Source: https://assist.dla.mil -- Downloaded: 20260-26T04:12Z
Check the source to verify that this is the current version before use.

2.1.2 Other Government documents and publications. The following other Government documents and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DEPARTMENT OF AGRICULTURE (USDA)

USDA Grading Standards

(Application for copies should be addressed to the US Department of Agriculture, Standards Section, Cotton Division, AMS-USDA, 4841 Summer Avenue, Memphis, TN 38122.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the non-Government documents which is current on the date of the solicitation.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

B2.1 - Pipe Threads (Except Dryseal) Specifications, Dimensions, and Gauging for Taper and Straight Pipe Threads, Including Certain Special Applications.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 380 - Cleaning and Descaling Stainless Steel Parts, Equipment and Systems.
A 484 - General Requirements for Stainless and Heat Resisting Bars, Billits and Forgings.
B 85 - Aluminum Alloy Die Castings.
B 211 - Aluminum Alloy Bar, Rod and Wire.
B 247 - Aluminum Alloy Die Forgings, Hand Forgings and Rolled Ring Forgings.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AEROSPACE MATERIAL SPECIFICATIONS (AMS)

5354 - Steel Castings, Investment Corrosion Resistant.
5678 - Steel Wire, Corrosion Resistant.
7455 - Bolt and Screws, Steel.

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

(Non-Government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:12Z
Check the source to verify that this is the current version before use.

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The sexless coupling assembly shall be either a dry break (valved) or a wet break (valveless) straight configuration with a 2- or 3-inch sexless end connection on one end and a 2- or 3-inch adapter end connection (sexless, male camlock, female camlock, external pipe thread, internal pipe thread, or replaceable hose) on other end; or fitting configurations (90-degree elbow, tee connection, Y connection, cross connection, double Y connection, emergency breakaway, closed circuit refueling nozzle, single point refueling fuel nozzle and pressure regulator) with 2- or 3-inch sexless end connections. The straight configuration coupling assembly shall consist of a coupling body, sexless end connection on one end; removable pressure caps secured by a wire cable to the body at the sexless end connection; sexless, male camlock, female camlock, external pipe thread, or replaceable hose end adapter connection on other end; and packing. Additionally, the valved configuration shall contain a valve with operating handle in each sexless end connection. The fitting configuration coupling assembly shall consist of a coupling body, sexless end connection(s), removable pressure cap(s) secured by a wire cable to the body at the sexless end connection(s), and packing. If coupling seals are used, the pressure cap shall contain a passive or active removable seal which can be interchanged with the coupling seal without the use of tools.

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 4.4 and 6.3). Any deviation of coupling assemblies from the approved first article during production, shall be subject to the approval of the contracting officer. Approval of the first article will not relieve the contractor of his obligation to furnish coupling assemblies conforming to this specification.

3.3 Materials. Materials selected by the contractor for those not specified shall be subject to all provisions of this specification. Materials shall be resistant to the fluid, fluid vapors, humidity, sand, solar radiation and other environmental factors specified and shall be of the lightest practicable weight. Copper or copper-bearing alloys shall not be used in any fuel-wetted parts.

3.3.1 Material deterioration prevention and control. The coupling assembly shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operating and storage enviornments to which the coupling may be exposed.

3.3.2 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:12Z
Check the source to verify that this is the current version before use.

3.3.3 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with component and subcomponent and shall make information available upon request to the contracting officer or designated representative.

3.3.4 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and re-processed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the coupling assembly may be newly fabricated from recovered materials to the maximum extent practicable, provided the coupling assembly produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the coupling assembly.

3.3.5 Aluminum components.

3.3.5.1 Forged components. Die and hand forged components shall be made of aluminum alloys conforming to ASTM B247 or MIL-A-12545 (impact forgings).

3.3.5.2 Cast components. Cast components shall be made of aluminum alloys conforming to ASTM B85 or MIL-C-11866.

3.3.5.3 Machined components. Machined components shall be made of aluminum alloys conforming to ASTM B211 or QQ-A-225/8.

3.3.5.4 Washers. Washers shall be made of aluminum.

3.3.6 Corrosion resistant steel components. Steel components shall be made of corrosion resistant steels conforming to ASTM A484, QQ-S-763 (Bar), AMS 5354 (Castings), AMS 5678 (17-7 PH Springs), AMS 7455 (440 Cres Balls) and austenitic Cres (Washers).

3.3.7 Wire cable. Wire cable shall be flexible wire rope, MIL-W-83420, type II, composition B.

3.3.8 Packing. Packing (seals and bumper) shall be made from MIL-R-25988 rubber or MIL-P-5315 fuel and oil resistant nitrile.

3.3.9 Plastic components. Plastic molded components shall be made from plastic that is resistant to the fluids and environmental factors specified herein.

3.4 Fluids. Coupling assembly components shall be compatible with MIL-F-46162, MIL-G-5572, MIL-T-5624, MIL-T-83133, and VV-F-800 fuels. In normal aircraft refueling operations these fuels may contain various quantities of dissolved and free water, salts, gases, some solid contaminants, such as sand, dirt and metal chips.

3.5 Design.

3.5.1 Valveless coupling assembly. The valveless coupling assembly shall conform to the design specified herein. The design shall be such that the mating parts of the coupling assembly shall provide a positive lock by

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:12Z
Check the source to verify that this is the current version before use.

satisfactorily engaging and sealing against any leakage during service use. A removable pressure cap shall be incorporated that will provide leak proof and dust proof connections with the sexless end. This removable pressure cap shall be attached to the coupling body with a wire long enough so the cap does not interfere with the fueling operation. Caps shall be connected to each other for protection of seal and to avoid contamination. The design shall also provide a 360-degree swivel feature to permit the orientation of the valve handle of valved coupling assembly when connected to a valved coupling. The design shall ensure electrical continuity during fuel operations by incorporating a dedicated contact. A protective shield shall be included that not only provides protection to seal out contaminants from entering when capped, during fueling operations, or to prevent damage when dropped or when coming in contact with surrounding surfaces. One man shall be capable of connecting or disconnecting the sexless end connection by hand.

3.5.2 Valved coupling assembly. The valved coupling assembly shall be designed to interface with the valveless coupling assembly. The valve shall prevent leakage in the closed position when the end cap is removed or when connected. The valve operating lever shall be conveniently located, however it shall be designed to prevent inadvertent valve opening or closing. The lever shall also be easily operable in any operating environment including quick shut-off capability while the system is pressurized and flowing. The valve lever shall be large enough to allow ease of operation by personnel wearing arctic mittens. The valve control shall be provided with a double-ended arrow showing direction of lever operation which is labled at each end to indicate the functional result (i.e., open, close, etc.). The design shall also provide a 360-degree swivel feature to permit the orientation of the valve handle of a valved coupling when connected to a valved or valveless coupling. The design shall ensure electrical continuity during fuel operations by incorporating a dedicated contact. A protective shield shall be included that not only provides protection to seal out contaminants from entering when capped, during fueling operations, or to prevent damage when dropped or when coming in contact with surrounding surfaces. One man shall be capable of connecting or disconnecting the sexless end connection by hand.

3.6 Construction. When tested as specified in 4.6.2.15, the lightweight coupling assembly shall be constructed to withstand the strains, jars, vibrations, and other conditions incident to service usage, shipping and storage.

3.7 Performance.

3.7.1 Operating pressure. The coupling assembly shall operate at pressures ranging from 12 inches mercury (Hg) vacuum to +100 psig with no evidence of binding, leaking, excessive force required to operate, or chattering.

3.7.2 Flow capacity. The coupling assembly pressure drop with a 200 gallons per minute (gpm) flow rate of fuel conforming to MIL-T-5624, grade JP-4 shall not be greater than listed below.

Coupling assembly nominal size, in.	Pressure drop, psi
2	3
3	.5

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:12Z
Check the source to verify that this is the current version before use.

3.8 Human factors engineering. The coupling assembly shall conform to Human Factors Engineering Design Criteria as described in MIL-STD-1472. Special design emphasis shall be given to: General Requirements, Controls, Labeling, Anthropometry, Design for Maintainability, and Hazards and Safety of MIL-STD-1472, as applicable.

3.9 Safety. The coupling assembly design shall incorporate safety features that will ensure personnel safety, minimize human error in the field, and prevent accidental spills.

3.9.1 Static control. To control static electricity during fueling operation, an electrical continuity device shall be incorporated in the coupling assembly. This device shall provide continuity contact between mating sexless surfaces without the use of external conductors. Provisions shall also be made to achieve intimate contact between hose bonding conductor and sexless coupling assembly when assembling the hose to the sexless coupling assembly.

3.9.2 Accidental spill prevention. Valve lever shall be designed to prevent inadvertent valve opening or closing.

3.9.3 Valve interlocks. A valve interlock shall be provided that prevents sexless connection or disconnection when valve is open.

3.10 Environmental requirements.

3.10.1 Operating temperatures. The coupling assembly shall perform as specified in 3.6 in any ambient temperature from +125 °F to -25 °F.

3.10.2 Storage temperature. The coupling assembly shall not be damaged by any ambient temperature from +155 °F to -65 °F.

3.11 Physical properties.

3.11.1 Hydrostatic pressure. The coupling assembly shall withstand an internal hydrostatic pressure of 200 pounds per square inch maintained for not less than 1 minute without damage or leaks.

3.11.2 Leakage. The copling assembly shall withstand internal hydrostatic pressures ranging from 12 inches of Hg, vacuum +5 and 100 psi with no damage leaks.

3.11.3 Spillage. Spillage from mated valved coupling assemblies when disconnected with valves closed shall not exceed 15 cubic centimeters of fluid for 2-inch couplings and 40 cubic centimeters for 3-inch couplings.

3.11.4 Swiveling torque. The torque required to swivel the coupling assembly shall not be greater than 240 inch-pounds.

3.11.5 Strength. The coupling assembly shall be sufficiently rugged to withstand normal operating, handling and maintenance loads without distortion or failure.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:12Z
Check the source to verify that this is the current version before use.

3.11.6 Valve actuation force. The force required to actuate the valve lever shall not exceed 15 pounds.

3.11.7 Electrical resistance. The coupling assembly electrical resistance shall not exceed 5 ohms at a potential of 1.5 volts.

3.11.8 Dust protection. The internal surfaces of the coupling assembly shall not be contaminated by blowing dust, sand, and other foreign particles when connected or disconnected and capped.

3.11.9 Corrosion. When tested in accordance with 4.6.2.12, the coupling assembly shall withstand the effects of high humidity and salt atmosphere.

3.11.10 Solar radiation. The coupling assembly shall not suffer loss of structural integrity when exposed to solar radiation.

3.11.11 Endurance. The coupling assembly shall be sufficiently rugged to withstand 2,000 connections and disconnections (complete cycles) when wet and 500 complete cycles when dry without functional distortion or failure.

3.11.12 Drop. The coupling assembly shall withstand rough handling by not suffering loss of structural integrity when dropped eight times from a height of 6 feet onto a concrete floor.

3.11.13 Burst pressure. The coupling assemblies when joined together shall withstand an internal hydrostatic pressure of not less than 300 pounds per square inch (psi) maintained for not less than 1 minute without damage or leaks.

3.12 Weight. The weight of the coupling assembly shall be as lightweight as practicable and still meet the requirements specified herein.

3.13 Color. Unless otherwise specified (see 6.2), the coupling assembly color shall be a nonvivid and lusterless dark shade of green except valve lever, caps and pivot pin if used shall be nonvivid and lusterless nonvivid and lusterless black.

3.14 Finish. The use of any protective coating that will crack, chip, peel, or scale is prohibited. Paint shall not be used on surfaces in contact with fuel.

3.14.1 Aluminum components. Aluminum components when used shall be anodized per MIL-A-8625, anodized to a thickness of 0.0007-inch for wrought aluminum alloys and 0.004-inch for aluminum castings alloys except in areas where continuity is required, and in those areas chemical film, per MIL-C-5541, shall be used.

3.14.2 Corrosion resistant steel components. Corrosion resistant steel components when used shall be passivated per ASTM A-380.

3.15 Identification marking. The coupling assembly shall be identified in accordance with MIL-STD-130.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:12Z
Check the source to verify that this is the current version before use.

3.16 Adapter end connections. Camlock connections shall conform to the requirements of MIL-C-27487. Hose connections shall be reusable, push-on clamp-type fittings and sized to mate with 2- or 3-inch standard collapsible or rigid hose.

3.17 Interchangeability. All parts having the same part number shall be functionally and dimensionally interchangeable. Interchangeable parts are defined as two or more like parts possessing such functional and physical characteristics as to be equivalent in performance and durability, and capable of being exchanged one for the other without alteration of the parts themselves or of adjoining parts without selection for fit or performance.

3.18 Workmanship. All parts, components, and assemblies of the couplings, including castings, forgings, stampings, and machined surfaces shall be clean and free from dust, pits, and dirt, sprues, scale, and other injurious defects. External surfaces shall be free of burrs, sharp edges, and corners except where sharp edges and corners are required or where they are not detrimental to safety.

3.18.1 Castings. Castings shall be sound and free from blowholes, patching, misplaced coring, warping, and any other defects which might render the castings unsound or unsuitable for their intended use.

3.18.2 Threads. Pipe threads shall be in accordance with ANSI B2.1.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Components and material inspection. The contractor is responsible for insuring that components and materials are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:12Z
Check the source to verify that this is the current version before use.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5).
c. Inspection of packaging (see 4.7).

4.3 Inspection conditions.

4.3.1 Ambient temperature. Unless otherwise specified in a test, test instruments, coupling assembly parts, and other devices used in dimensioning, gauging, and performance testing shall be stabilized at, and used in, an ambient temperature of 75 ±15 °F.

4.3.2 Barometric pressure. Barometric pressure shall be 14.7 ±1 pound per square inch.

4.3.3 Ambient humidity. Ambient humidity shall not be greater than 90 percent relative humidity.

4.3.4 Other condition tolerances. Unless otherwise specified, other condition tolerances shall be:

a. Temperature - ±4 °F
b. Pressure - ±5 percent
c. Humidity - ±5 percent
d. Time - ±2 percent of elapsed time

4.4 First article inspection.

4.4.1 First article examination. The first article shall be examined as specified in 4.6.1, in the order shown in table I. The presence of one or more defects shal be cause for rejection.

4.4.2 First article tests. The first article shall be tested as specified in 4.6.2 and in column 1 of table II. Failure of any test shall be cause for rejection. Coupling assembly accessories such as camlock adapters and hoses need not be subjected to the testing required in 4.6.2.2, 4.6.2.4, 4.2.6.7, 4.6.2.11, 4.6.2.17 and 4.6.2.18.

4.5 Quality conformance inspection.

4.5.1 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.5.2 Examination. Samples selected in accordance with 4.5.1 shall be examined as specified in 4.6.1. AQL shall be 1.0 percent defective for major defects.

4.5.3 Tests.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:12Z
Check the source to verify that this is the current version before use.

4.5.3.1 Individual. Each coupling assembly shall be subjected to the tests marked "X" in column 2 of table II. Failure any test shall be cause for rejection.

4.5.3.2 Samples. Samples selected in accordance with 4.5.1 shall be subjected to the tests marked "X" in column 3 of table II. AQL shall be 1.0 percent defective.

4.6 Examination procedure.

4.6.1 Examination. The coupling assemblies shall be examined as specified herein for the following defects listed in table I.

TABLE I. Examination schedule.

Number	Defects	Requirement paragraph
	Major	
101.	Material not as specified.	3.3
102.	Materials not resistant to corrosion and deterioration or treated to be made resistant to corrosion and deterioration.	3.3.1
103.	Dissimilar metals not in accordance with MIL-STD-889.	3.3.2
104.	Contractor does not have documentation available for identification of material, material finishes, or treatment.	3.3.3
105.	Used, rebuilt, or remanufactured components, pieces, or parts incorporated.	3.3.4
106.	Valve not functioning properly.	3.7.1
107.	Noncompliance to environmental conditions.	3.10
108.	Threads not as specified.	3.18.2
109.	Camlock fitting not as specified.	3.16
110.	Hose connection not as specified.	3.16
111.	Color not as specified.	3.13
112.	Treatment and finish not as specified.	3.14
113.	Identification marking missing, incomplete or illegible.	3.15
114.	Noncompliance to safety conditions.	3.9
115.	Noncompliance to human factors engineering.	3.8
116.	Cleaning not as specified.	3.18
117.	Workmanship not as specified.	3.18
118.	Aluminum components not as specified.	3.3.5
119.	Forged components not as specified.	3.3.5.1
120.	Die cast components not as specified.	3.3.5.2
121.	Machined components not as specified.	3.3.5.3

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:12Z
Check the source to verify that this is the current version before use.

TABLE I. Examination schedule. (Cont'd)

Number	Defects	Requirement paragraph
122.	Corrosion resistant steel components not as specified.	3.3.6
123.	Wire cable not as specified.	3.3.7
124.	Packing not as specified.	3.3.8
125.	Plastic components not as specified.	3.3.9
126.	Design not as specified for valveless assembly.	3.5.1
127.	Design not as specified for valved assembly.	3.5.2
128.	Valve handle cannot be accidently opened or closed as specified.	3.9.2
129.	Valve interlocks not as specified.	3.9.3
130.	Weight not as low as possible.	3.12
131.	Castings not as specified.	3.18.1
132.	Interchangeability not as specified.	3.17

TABLE II. Test schedule.

First article 1		Quality conformance			Test 4	Test Paragraph 5	Requirement Paragraph 6
		Individual 2	Sample 3				
Valve	Valveless		Valve	Valveless			
X	X	X	-	-	Hydrostatic pressure	4.6.2.2	3.11.1
X	X	-	X	X	Functional	4.6.2.3	3.11.1
					Leakage	4.6.2.4	3.11.2
X	X	-	X	X	Connected	4.6.2.4.1	
X	-	-	X	-	Disconnected	4.6.2.4.2	
X	-	-	-	-	Spillage	4.6.2.5	3.11.3
X	X	-	-	-	Pressure drop	4.6.2.6	3.7.2
X	X	-	-	-	Strength	4.6.2.7	3.11.4
X	X	-	-	-	Swiveling torque	4.6.2.8	3.11.5

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:12Z
Check the source to verify that this is the current version before use.

TABLE II. Test schedule.

First article 1		Quality conformance			Test	Test Paragraph	Requirement Paragraph
		Individual	Sample 3				
Valve	Valveless	2	Valve	Valveless	4	5	6
X	-	-	-	-	Valve actuation force	4.6.2.9	3.11.6
X	X	X	-	-	Electrical resistance	4.6.2.10	3.11.7
X	X	-	-	-	Dust	4.6.2.11	3.11.8
X	X	-	-	-	Accelerated corrosion	4.6.2.12	3.4/3.11.9
X	X	-	-	-	Solar radiation	4.6.2.13	3.11.10
X	X	-	-	-	Fuel resistance and extreme temperature	4.6.2.14	3.3/3.9.1
X	X	-	-	-	Contaminated fuel	4.6.2.15	3.6
					Endurance	4.6.2.16	3.11.11
X	-	-	-	-	Wet	4.6.2.16.1	
X	X	-	-	-	Dry	4.6.2.16.2	
X	X	-	-	-	Drop	4.6.2.17	3.11.12
X	X	-	X	X	Burst pressure	4.6.2.18	3.11.13
X	X	-	X	X	Disassembly and inspection	4.6.2.19	3.2

4.6.2 Tests.

4.6.2.1 Test conditions.

4.6.2.1.1 Cleaning. Prior to testing test samples, all internal surfaces normally in contact with the fluid shall be thoroughly cleaned to remove foreign matter.

4.6.2.1.2 Test fluid. Unless otherwise specified, the test fluid shall be water or solvent conforming to MIL-C-7024.

4.6.2.2 Hydrostatic pressure. A coupling assembly mated to another coupling assembly (mated coupling assembly) shall be subjected to a hydrostatic pressure

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:12Z
Check the source to verify that this is the current version before use.

of 200 pounds per square inch for not less than 1 minute. Any damage or leak shall be cause for rejection. The valved mated coupling assembly shall be tested twice, with valves open and valves closed and capped.

4.6.2.3 Functional test. The valved mated coupling assembly shall be installed in a flow test stand and functionally tested. Before starting fluid flow, the valve(s) in the coupling assemblies shall be opened and closed at least 10 times. Then an attempt shall be made to disconnect the mated coupling assembly sexless end connection. The coupling interlocks shall prevent disconnection. With valves open, fluid shall be allowed to flow through with flows ranging from 0 to 200 gallons per minute in 50-gallon per minute increments. At each flow increment, each valve shall be closed and opened not less than 10 times without evidence of binding, of excessive force being required, or of chattering. Nonconformance to 3.11.1 shall constitute failure of this test.

4.6.2.4 Leakage.

4.6.2.4.1 Connected position. The connected coupling assembly shall be subjected to internal hydrostatic pressures of 12 inches Hg vacuum, +5 and +100 psi. The pressure shall be maintained at each increment for not less than 1 minute. There shall be no loss of vacuum during negative pressure or evidence of leakage during positive pressure.

4.6.2.4.2 Disconnected position. Only the valved coupling assembly with valve closed shall be subjected to the internal hydrostatic pressure test specified in 4.6.2.4.1. Nonconformance to 3.11.2 shall constitute failure of this test.

4.6.2.5 Spillage test. The valved mated coupling assembly shall be subjected to a fluid flow rate of not less than 50 gallons per minute. First the downstream coupling assembly valve shall be closed, then the upstream coupling assembly valve shall be closed, and the coupling assembly sexless matings ends disconnected. Combined fluid spillage at the sexless mating connections during disconnection greater than 15 cubic centimeters shall be cause for rejection for 2-inch couplings and 50 cubic centimeters for 3-inch couplings. Nonconformance to 3.11.3 shall constitute failure of this test.

4.6.2.6 Pressure drop. The mated coupling assembly (valved with valve open) shall be installed in a flow test stand and shall be tested for pressure drop in one direction with a flow of 200 gallons per minute. The pressure drop shall not be greater than specified in 3.7.2 with a 200 gallon per minute flow rate of fuel conforming to MIL-T-5624, grade JP-4 at test temperature. If test fluid used differs from referenced fluid, the results should be corrected using the formula of MIL-C-81975 as follows:

$$PD_{(corr)} PD_{(net)} \times \frac{(V)}{V_t} \quad .025 \quad \times \frac{SG}{SG_t}$$

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:12Z
Check the source to verify that this is the current version before use.

Where V = absolute viscosity of referenced fluid in centistokes
V_t = absolute viscosity of test fluid in centistokes
SG = specific gravity of referenced fluid
SG_t = specific gravity of test fluid
$PD_{(net)}$ = pressure drop obtained with test fluid in psi
$PD_{(corr)}$ = pressure drop corrected to referenced fluid in psi

Nonconformance to 3.7.2 shall constitute failure of this test.

4.6.2.7 Strength. The coupling assembly shall be subjected to a 200 foot-pound bending force applied at one end (in a direction 90 degrees to the longitudinal centerline) while the other end is pressurized to 100 pounds per square inch internal pressure. While maintaining the pressure, the load shall be applied and removed 100 times. There shall be no leakage during this test or damage which causes malfunctioning of the assembly as a result of this test. The test shall also be repeated at a vacuum of 12 inches of mercury. At the conclusion of this test, the assembly shall be subjected to a leakage test as specified in 4.6.2.4. Nonconformance to 3.11.4 shall constitute failure of this test.

4.6.2.8 Swiveling torque. The coupling assembly shall be rotated 360 degrees in each direction to measure the starting torque when the internal pressure is 0 and 100 pounds per square inch. Torque required to start or to sustain rotation shall not be greater than 240 inch-pounds. Nonconformance to 3.11.5 shall constitute failure of this test.

4.6.2.9 Valve actuation force. The valved coupling assembly valve shall be opened and closed when the internal pressure is 0 and 100 pounds per square inch. The force required to open or close the valve shall not be greater than 15 pounds.

4.6.2.10 Electrical resistance. The valved coupling assembly electrical resistance between two mated coupling assemblies shall be measured. The electrical resistance shall not be greater than 5 ohms when a potential of 1.5 volts is applied.

4.6.2.11 Dust. The dust test shall be performed in accordance with MIL-STD-810, method 510.1. Immediately after this test, the assembly shall be inspected and there shall be no evidence of deterioration which might adversely affect subsequent operation. The assembly shall then be subjected to the valve actuation force test in 4.6.2.9, swiveling torque test in 4.6.2.8 and leakage test in 4.6.2.4. Nonconformance to 3.11.8 shall constitute failure of this test.

4.6.2.12 Accelerated corrosion. The coupling assembly with all fuel ports open, shall be submerged in a 2.5 percent solution by weight of sodium chloride and water. The assembly shall then be drained for 30 seconds and immediately placed in a test chamber maintained at a temperature of 86 °F with a relative humidity of 100 percent for a period of 20 minutes. Upon completion of the humidity exposure period, the unit shall be placed in an air oven maintained 130 °F for a period of 20 minutes. The assembly shall not be operated again until a total of 50 immersion, high humidity and drying cycles have been

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:12Z
Check the source to verify that this is the current version before use.

completed. Immediately after th 50th cycle, the assembly shall be washed with warm water to remove all salt accumulations and then dried. There shall be no evidence of deterioration which might adversely affect subsequent operation. The valve assembly shall then be subjected to the valved actuation force test in 4.6.2.9, to the swiveling force test in 4.6.2.8 and to the leakage test in 4.6.2.4.

4.6.2.13 Solar radiation. The solar radiation test shall be performed in accordance with MIL-STD-810, method 505.2. Nonconformance to 3.11.10 shall constitute failure of this test.

4.6.2.14 Fuel resistance and extreme temperature.

4.6.2.14.1 Low temperature. The coupling shall be filled with fuel conforming to TT-S-735, type I and then shall be placed in a low temperature chamber and maintained at -65 ±2 °F for 24 hours after the chamber temperature is stabilized. Upon completion of the 24-hour soak the coupling assembly shall be inspected for damage and degradation. The coupling assembly shall then be soaked at -25 ±2 °F for 24 hours after the chamber temperature has stabilized. After completion of the 24-hour soak, while at -25 °F the assembly shall be swiveled in each direction 10 times. On valved assemblies the valve shall be opened and closed 10 times. Afterwards the assembly shall be subjected to the leakage test in 4.6.2.4.

4.6.2.14.2 High temperature. The coupling assembly shall be filled with and immersed in a container filled with fuel conforming to MIL-T-5624, grade JP-5 or JP-8 and then shall be placed in an oven maintained at 155 ±2 °F. To prevent boiling, it is advised that the container be closed with pressure maintained no greater than 15 pounds psi. After the oven temperature is stabilized, the assembly shall be soaked for 72 hours. After the completion of the 72-hour soak, the assembly shall be removed from the oven and air dried for 72 hours. The assembly shall then be subjected to the dry endurance test in 4.6.2.16.2.

4.6.2.14.3 Failure criteria. Nonconformance to 3.9.1 shall constitute failure of these tests.

4.6.2.15 Contaminated fuel test. The mated coupling assembly shall be subjected to 200 gallons per minute flow at 100 pounds per square inch pressure for 2.5 hours and to 10 to 25 percent rated flow and pressure for 2.5 hours and shall be repeated once. The test fuel contaminant mixture shall conform to that shown in table III. At the completion of this test, the valved assembly shall be subjected to the valve operating force in 4.6.2.9, to the swiveling torque test in 4.6.2.8, and the leakage test in 4.6.2.4.

4.6.2.16 Endurance.

4.6.2.16.1 Wet. The valved mating coupling assembly shall be connected and disconnected in the normal manner 2000 times (complete cycle) while fuel pressure of 100 pounds per square inch is applied internally each time the connection is made. The valve operating force test in 4.6.2.9 and swiveling torque test in 4.6.2.8 shall be conducted after each 200 cycles. At the completion of this test, the leakage test in 4.6.2.4 shall be conducted. One test cycle shall consist of:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:12Z
Check the source to verify that this is the current version before use.

a. Connect two coupling assemblies together.
b. Open coupling assembly valves.
c. Begin flow.
d. Rotate (swivel) alternately in each direction.
e. After not less than 10 seconds of flow, close coupling assembly valves.
f. Disconnect assembly.

TABLE III. Contaminant mixture.

CONTAMINANT	PARTICLE SIZE (Microns)	QUANTITY (GRAMS per 1000 liters)
Iron Oxide	0 - 5 5 - 10	19 1.0
Sharp Silica Sand	150 - 300 300 - 420	0.7 0.7
Prepared dirt conforing to AC Spark Plug Co. Part No. 1543637 (Coarse Arizona road dust) or equal	Mixture as follows: 0 - 5 (12%) 5 - 10 (12%) 10 - 20 (14%) 20 - 40 (23%) 40 - 80 (30%) 80 - 200 (9%)	0.3
Cotton linters	Staple Below 7 U.S. Dept of Agriculture Grading Standards	0.07
Iron Chips	150 - 500	10
Aluminum Chips	150 - 500	10

4.6.2.16.2 Dry. The dry endurance test shall be conducted in a manner similar to the wet endurance test, except that the duration of the test shall be 500 cycles under dry conditions. The valve operating force test in 4.5.2.9 and swiveling torque test in 4.6.2.8 shall be conducted upon completion of the dry endurance test. Following the swiveling torque test, the leakage test in 4.2.6.4 shall be conducted. One cycle shall consist of:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:12Z
Check the source to verify that this is the current version before use.

a. Connect two couplings assemblies together.
b. Rotate (swivel) alternately in each direction.
c. Open and close valves.
d. Disconnect assembly.

4.6.2.17 Drop. The mated coupling assembly that is capped at both ends shall be dropped from a height of 6 feet onto a concrete floor. Drop the mated coupling assembly 2 times in each position listed below.

a. Valveless - Coupling axis horizontal, top
- on side, rotated 90° from top position
- on bottom, rotated 90° from bottom position
- on dust cap

b. Valved - coupling axis horizontal, valve on top
- coupling axis horizontal, valve on side
- coupling axis horizontal, valve on bottom
- on dust cap

The valveless assembly shall then be subjected to the swiveling torque test in 4.6.2.8 and the leakage test in 4.6.2.4. The valved assembly shall then be subjected to the valve actuation force test in 4.6.2.9, the swiveling torque test in 4.6.2.8, and the leakage test in 4.6.2.4. There shall be no leakage or damage which causes malfunctioning as a result of the drop test.

4.6.2.18 Burst pressure test. The coupling assembly shall be subjected to a burst pressure of 300 pounds psi for not less than 1 minute. Any damage or leaks shall be cause for rejection.

4.6.2.19 Disassembly. Upon completion of testing, all coupling assemblies shall be disassembled and visually inspected. There shall be no evidence of excessive wear, deterioration or corrosion which could cause malfunctioning.

4.7 Inspection of packaging.

4.7.1 Quality conformance inspection.

4.7.1.1 Unit of product. For purposes of inspection the following shall apply:

a. Unit pack - A preserved coupling in its unit pack container.
b. Shipping container - A completed shipping container ready for shipment.

4.7.1.2 Sampling. Sampling of unit packs shall be in accordance with MIL-STD-105. The AQL shall be 1.0 percent defective.

4.7.1.3 Examination.

a. Samples of unit packs selected in accordance with 4.7.1.2 shall be examined for the following major defects:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:12Z
Check the source to verify that this is the current version before use.

118. Protective caps not in place for level A and C (see 5.1.1 and 5.1.2).
119. Preservative not applied as specified for level A (see 5.1.1).
120. Barrier material not of type specified or secured as specified for level A (see 5.1.1).
121. Unit pack container not of type specified for level A and C (see 5.1.1, 5.1.2).
122. Marking not as specified for level A and C (see 5.3).

b. Each shipping container shall be examined for the following major defects:

123. Shipping containers not as specified for levels A, B, and C (see 5.2.1, 5.2.2, 5.2.3).
124. Weight of shipping containers exceeds amount specified for levels A, B, and C (see 5.2.1, 5.2.2 and 5.2.3).
125. Strapping not as specified for level A (see 5.2.1)
126. Marking missing, incorrect, or illegible for levels A, B, and C (see 5.3).

5. PACKING

5.1 Preservation. Preservation shall be level A or C as specified (see 6.2).

5.1.1 Level A. Protective caps shall be in place to prevent entrance of dirt into the coupling. Unpainted, exterior steel surfaces shall be coated with P-19 preservative as specified in MIL-P-116; preservative shall not be applied to seals, gaskets, plastic parts, or the interior of the coupling. The coupling shall be wrapped with barrier material conforming with MIL-B-121, grade A, type II, class 2 secured in place with tape conforming to PPP-T-60, type optional. Each coupling shall be unit packed in a fiberboard box conforming to PPP-B-636, type CF, class weather-resistant, grade W6c; the box shall be closed and sealed using method IV in accordance with the appendix to the box specification.

5.1.2 Level C. Protective caps shall be in place. Each coupling shall be unit packed in a box conforming to PPP-B-636, type CF, class domestic, grade optional. The box shall be closed in accordance with the box specification.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. Couplings, preserved as specified in 5.1, shall be packed in a box conforming to PPP-B-601, overseas type, style optional. Gross weight of the container shall not exceed 150 pounds. Strapping shall be in accordance with box specification except that it shall be flat steel, finish B.

5.2.2 Level B. Couplings shall be packed in a box conforming to PPP-B-636, type CF, class weather-resistant, varity SW or DW; the load shall not exceed the weight or size limits specified for the grade selected. The box shall be closed, sealed, and reinforced in accordance with the appendix to the box specification.

5.2.3 Level C. Couplings shall be packed in domestic grade fiberboard boxes in quantities not exceeding the weight and size limitations of the box. The box shall be closed and sealed with tape or adhesive.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:12Z
Check the source to verify that this is the current version before use.

5.3 Marking. Marking of unit and shipping containers shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The sexless type quick-disconnect coupling assemblies covered by this specification are intended for use in aircraft fuel servicing systems.

6.2 Order data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Coupling assembly requireed as specified on acquisition documents (see 1.2).
c. Time frame required for submission of first article coupling assemblies (see 3.2).
d. When the Government will conduct some but not all of the first article examination and tests, the contracting officer should specify which examination and test will be conducted by the Government and which examination shall be conducted by the contractor (see 3.2 and 4.4).
e. When color shall not be a dark shade of green (see 3.13).
f. When certification will be accepted in lieu of verification (see 4.4).
g. Levels of preservation and packing required (see 5.1 and 5.2).

6.3 First article. When a first article inspection is required, the items should be initial production. The first article should consist of one or more coupling assemblies. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, tests, and approval of the first article test results and disposition of the first article.

6.4 Subject term (key word) listing.

Aircraft, fuel servicing system
Coupling assembly
Quick disconnect coupling
Sexless type

Custodian:

Army - ME

Preparing activity:

Army - ME

Project 4730-A022

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:12Z
Check the source to verify that this is the current version before use.

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-53071(ME)	Coupling Assemblies, Quick Disconnect, Sexless Type

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:12Z
Check the source to verify that this is the current version before use.